Department of Information Technology
Academic Year 2023 - 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT
Course Code & Title: CS3452 & Theory of Computation
Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V / P and NP completeness
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO18
- **Activity Chosen:** Learning by Teaching
- **Justification:**
  - In unit V, the students need to understand the P and NP completeness problems. This activity helps the students to understand the topic and increases the leadership skills of a student.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
  - A topic P and NP completeness problems was given to Mr.S. Sarukumar and Ms. S.Sobana one week before the presentation.
  - He prepared the presentation and taken the class on 31.05.2024.
  - The topic was delivered around 20 minutes with an example problem.
  - The students discussed with an example using the different attributes and styles.
  - The implementation of Learning by teaching activity as shown in figure1 and figure2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO5	2	3	1	2	2	1	2	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO8**	**PO9**	**PO10**	**PO12**	**PSO1**	**PSO2**	**PSO3**
	2	3	1	2	2	1	2	2	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of P and NP completeness	The students will be able to identify the need for P and NP completeness of complex engineering problems	Able to design solutions for any applications using P and NP completeness	Able to analyze and apply P and NP completeness	Able to follow ethics for solving P and NP completeness	Able to implement the problem individually	Able to implement P and NP completeness	Ability to recognize the need for P and NP completeness in real world applications	Exhibit knowledge in P and NP completeness for application development	Exhibit knowledge in P and NP completeness in TOC of IoT, Cloud Computing and Data Analytics to provide IT solutions	Exhibit knowledge in P and NP completeness in software projects to deliver a quality product.

- **Images / Screenshot of the practice:**

Figure 1: Learning by Teaching Handled by Sri Sarukumar S

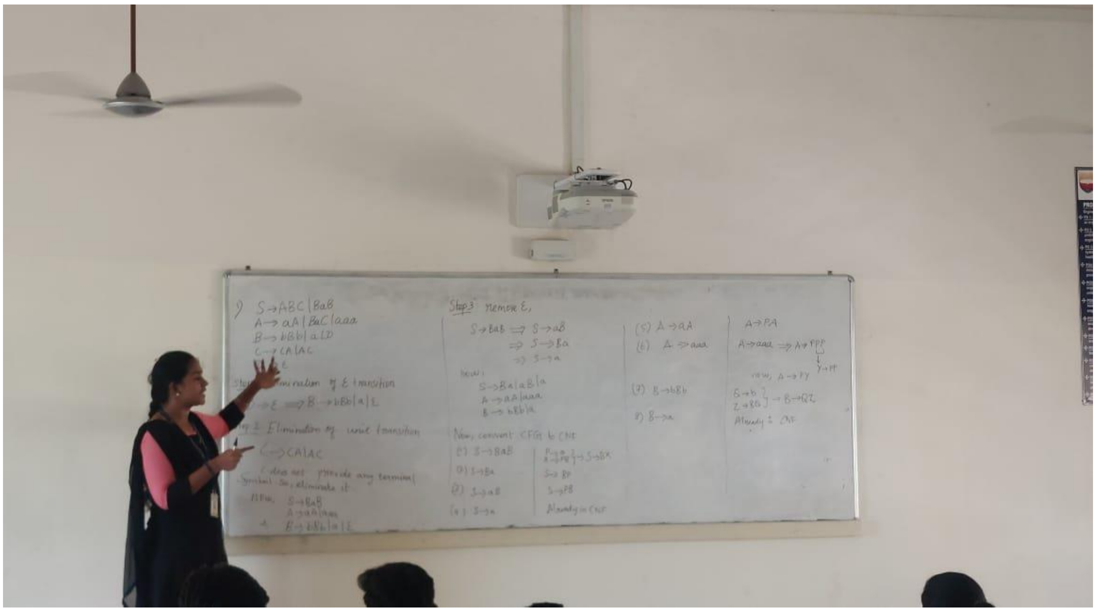

Figure 2: Learning by Teaching

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - The students felt that they are able to understand the topic as it was taught by their classmate
    - They felt comfortable to ask doubts while teaching.
    - The student who took the class said it helps in improving self learning skills
  - ***Benefit of the practice:***
    - The student's technical knowledge, confidence, and communication improve as well as they teach the new subject.
    - Peer-to-peer instruction helps students develop strong communication skills.
    - Students are able to impart their newly learned knowledge.

❖ ***Challenges faced in implementation:***

- Effectively encourage students who are not engaging in the activity by highlighting the advantages of self-learning or autonomous learning.
- Unable to convince all students to participate in self-learning because the activity only includes one student.

References:

❖ https://ohioline.osu.edu/

❖ https://www.nureva.com/blog/education/15-active-learning-activities-to-energize-your-next-college-class

❖ https://www.ritrjpm.ac.in/images/computer-science/

Signature of Faculty Member **HOD**